MIL-D-45089A(Ord)
30 April 1962
SUPERSEDING
MIL-D-45089(Ord)
5 June 1958

MILITARY SPECIFICATION

DRIVE-MOW, PERISCOPE: M9

1. SCOPE

1.1 This specification covers one particlar type of instrument, performing the functions of the conventional ballistics drive and the periscope mount, hereinafter referred to as drive-mount.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitatlion for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-515	Plywood; Container Grade
UU-T-106	Tape, Masking; Paper (Pressure Sensitive)
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof - For Packaging and Sealing
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

FSC 1290

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Military

MIL-C-104	Crates, Wood; Lumber and Plywood, Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-P-130	Paper, Wrapping, Laminated and Creped
MIL-B-131	Barrier Material, Water-vaporproof, Flexible
MIL-L-3150	Lubricating Oil, Preservative, Medium
MIL-A-5092	Adhesive, Rubber, (Synthetic and Reclaimed Rubber Base)
MIL-R-6855	Rubber; Synthetic Sheet, Molded, and extruded, for aircraft applications
MIL-C-11796	Corrosion Preventive, Petrolatum, Hot Application
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45208	Inspection Requirements, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling, Procedures and Tables for Inspecting by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8606140	Drive-Mount, Periscope: M9
LSQAP-8606140	List of Supplementary Quality Assurance Provisions for Drive Mount, Periscope: M9

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one drive-mount, representative of production, shall be made available to the Government to determine if the drive-mount meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction. - The drive-mount shall be manufactured in accordance with Drawirg F8606140, "Drive-Mount, Periscope: M9" and drawings, pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Operating temperatures.- The drive-mount shall meet the applicable requirements of 10, 3.11.1, 3.12 and 3.13.1 while exposed to and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit.

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3.5.2 Vibration.- The drive-mount complete, including shield, headrest and linkages shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second at an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration the drive-mount shall show no evidence of damage, loose or missing parts and shall meet the requirements of this specification.

3.6 Orientation.- The drive-mount shall be complete, including the shield headrest and linkages, when determining compliance with reguirements of this specification. The locating surface of the mounting flange of the mount shall be secured in a horizontal plane. The index line on the input shaft shall be alined to the arrow on the qualification plate for zero superelevation in accordance with the value and direction indicated thereon. The inner and outer scales of the eccentric adjustment on the linkage arm shall be set to “0” . The stud in the free end of the parallelogram linkage shall be secured in approximately the same position as it would be when assembled to the gun trunnion bracket. The 14 inch Arm Assembly 8606131 shall be positioned to the 9.899 inch dimension in accordance with Drawing F8606140-1. The level vial shall be central within the chickness of one raduation line. The free end of the compensating Link Assembly 8606045 shall be secured in the approximate position shown on Drawing F8606140-1.

3.7 Superelevation input travel.- Rotation of the input shaft to the extreme clockwise and counterclockwise positions shall be limited only by the stops and there shall be no evidence of interference or jamming as a result of “stop” action. A minimum of 22 full turns of the input shaft shall be available over the entire range of rotation.

3.8 Superelevation accuracy.- Rotation of the superelevation input shaft 5 revolutions counterclockwise, plus or minus 30 minutes,backlash excluded, shall result in the Output Coupling 1298344 rotating 24.98 roils plus or minus 0.1 mil. Continuation of rotation to 10 revolutions, plus or minus 30 minutes, backlash excluded, shall result in the coupling rotating 49.96 mils plus or minus 0.l mil.

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3.9 Backlash.

3.9.1 Superelevation backlash.- With a 10 pound inch load applied at the output coupling, backlash in the superelevation mechanism at the output coupling shall not exceed 0.1 mil as a result of change in direction of rotation of the superelevation input shaft. Backlash at the 5 and 10 revolutions counterclockwise positions of the input shaft shall not exceed 0.1 mil.

3.9.2 Elevation backlash.- With a 10 pound inch load applied at the output coupling, backlash shall not exceed 0.1 mil at the output coupling when elevation and depression movements are introduced at the simulated gun trunnion bracket. Backlash at simulated gun elevations of approximately 8 degrees and 15 degrees elevation shall not exceed 0.1 mil.

3.10 Shield assembly.- Actuation of the handle shall operate the shield when the plunger is depressed. The handle assembly shall remain locked in the open and closed positions when the plunger is engaged.

3.11 Periscope holding features.- An acceptable Periscope: M29, or the equivalent there of, shall be utilized for insertion and removal in the drive-mouni to determine the following functional operations of the Plate Assembly 8298431.

3.11.1 Release grip.- When the periscope is inserted into the mount, the mount shall automatically latch the periscope in the fully-inserted position so that it cannot be removed without pulling on the lever of Lever Assembly 8298434. A direct pull force of not more than 12 pounds on the lever shall free the periscope and permit it to be withdrawn without restriction. The pull force on the lever assembly recorded at standard ambient temperature shall not change by more than 3 pounds when the mount is subjected to the temperatures specified in 3.5.1.

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3.11.2 Lever.- When the plunger for the Lever Assembly 8298434 depressed, the lever moved to its maximum forward position, and the plunger released, the lever shall secure the periscope in the mount so that a pressure of 10 pounds applied at the lowest point of the body assembly of the periscope in either direction (front and rear) shall not cause the periscope line of sight to change by more than .05 mil. When the plunger is released, the plunger spring action shall secure the lever in the locked position.

3.12 Headrest.- Detent action shall be posttive at both ends of the slide adjustment and the headrest shall slide smoothly without interference when pushed right and left between the detent positions.

3.13 Operability.

3.13.1 Spindle arm.- The torque required to rotate the spindle of Arm Assembly 8602723 shall not exceed 3 pound inches at standard ambient temperature, plus 60 degrees to 90 degrees Fahrenheit. The value recorded at standard ambient temperature shall not change in excess of one pound inch when torque is measured at the extreme temperatures of 3.5.1.

3.13.2 Eccentric.- The torque required to rotate the eccentric shall not exceed 4 pound inches at standard ambient temperature.

3.14 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. **QUALITY ASSURANCE PROVISIONS**

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 General requirements.- Inspection shall be in accordance with General Specification MIL-I-45208.

4.3 Production model.- Unless otherwise specified in the contract, one drive-mount shall be selected by the Government inspector as soon as practicable after the start of production and shall be inspected by the contractor under the surveillance of the Government. This i nspection shall be conducted at the contractor's facility, or Government approved laboratory using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined-and tested by the contractor under the surveillance of the Government. Acceptance of production lots wi1l not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 Defective units and lots: Defective units and lots shall be processed as specified in standard MIL-STD-105.

4.4.2 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP) listed in Supplementary Quality Assurance Provisions List (LSQAP)-8606140.

4.4.2.1 Contents of SQAP.- The contents of each SQAP shall represent the minimum inspectlon requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications. and to resect individual non-conforming items.

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4.4.2.2 Contractor's inspection for all components not covered by SQAP.- Inspection for these components shall be in accordance with the inspection procedures of the written inspection system required by Specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements and have a bearing on the functional accuracy and performance of the drive-mount.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified and are in addition to the accuracies listed below. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.2 Inspection equipment design requirements.

4.6.3 Device to test orientation, superelevation input travel; superelevation accuracy, superelevation backlash, and periscope holding lever.

4.6.3.1 The device shall be equipped with suitable means to permit cross-leveling the mounting flange locating surface in a horizontal plane within thirty seconds. Provisions for each drive-mount being tested to maintain the 9.899 plus or minus .001 dimension with a simulated gun trunnion holding the Spindle C8602698 and a surface to secure the free end of the compensating linkage in position as required by and depicted on Drawing F8606140.

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4.6.3.2 Provisions shall be made to permit rotation of twenty-two 22 revolutions, clockwise and counterclockwise, of Shaft C8606024. A measuring device, capable of measuring rotation of Coupling F8298344 from zero to at least fifty mils, graduated in one second increments, the accuracy of which shall be within nine seconds throughout the entire excursion, shall be incorporated.

4.6.3.3 A suitable means shall be provided to permit positioning the simulated gun trunnion within plus or minus 30 minutes at zero and 8 degrees elevation and applying a load of ten pound-inches plus or minus one-half pound-inch on Coupling F8298344 while testing requirements of superelevation backlash and elevation backlash.

4.6.3.4 Provisions shall be made to provide an appropriate target to be utilized for measuring the movement of the periscope line of sight. Acceptance and rejection limitation markings or pips shall be marked thereon, the angular tolerance of which shall not exceed five seconds. Rigidity, stability, ease of maintenance and adaptability of set-up and/or calibration equipment with complete provisions are also prime requirements. All special set-up and/or calibration equipment necessary for certification of device accuracy shall be furnished and will become a part of the device.

4.6.4 Device to test vibration requirements.- The device shall consist of a type of vibrating machine capable of producing vertical motion at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion), and shall be equipped with a medium capable of holding the drive-mount securely to the vibrating machine. The medium shall hold the drive-mount in a position as depicted on Drawing F8606140. The base of the medium shall be secured firmly to the vibrating machine.

4.6.5 Device to test torque requirements.- A torque wrench type tester with a minimum range of from zero pound-inches to and including six pound-inches, the accuracy of which shall be plus or minus one-quarter pound-inch throughout the entire zone of excursion, shall be provided. The tester shall be equipped with suitable adapters to accommodate testing of torque of spindle of Arm D8602723, superelevation input shaft and spindle of eccentric.

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4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables I, II, III, and IV shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	**Req't Par.**	**QA. Par.**	**Test Method**
101. Orientation	3.6	4.8.3	Device, Special Testing (DST)-Visual
102. Superelevation Input travel	3.7	4.8.4	DST
103. Superelevation accuracy	3.8	4.8.5	DST
104. Superelevation backlash	3.9.1	4.8.6	DST
105. Elevation backlash	3.9.2	4.8.7	DST
106. Lever function	3.11.2	4.8.9.2	DST-Functional
107. Torque, arm spindle	3.13.1	4.8.11	DST
108. Torque, eccentric spindle	3.13.2	4.8.12	DST
MINOR: AQL 4.0%			
201. Shield assembly function	3.10	4.8.8	DST-Functional
202. Release grip function	301101	4.8.9	DST-Functional
203. Headrest movement	3.12	4.8.10	Tactile
202. Identification marking	3.14	4.8.13	Visual

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TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L8, and Sampling Plan Table IV-A of Standard MIL-STD-305

CRITICAL: NONE DEFINED

MAJOR: NONE DEFINED

MINOR: AQL 4.0%	**Req't Par.**	**QA. Par.**	**Test Method**
201. Shield assembly function at extreme temperatures	3.5.1 3.10	4.8.1	Standard Measuring Equipment (SME) Functional
202. Release grip function at extreme temperatures	3.5.1 3.11.1	4.8.1	SME-Functional
203. Headrest movement at extreme temperatures	3.5.1 3.12	4.8.1	SME-Tactile
204. Torque, arm spindle at extreme temperatures	3.5.1 3.13.1	4.8.1	SME
205. Vibration	3.5.2	4.8.2	SME

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 2.5%	**Req't Par.**	**Test Method**
101. Link, compensating tube, and headrest assemblies disassembled (Level A,C)	5.3.1.2.1 5.3.1.1.2 5.3.1.1.3 5.3.2.1	Visual

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TABLE III - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 2.5% (cont'd)	**Req't Par.**	**Test Method**
302. Cleaning Inadequate or improper (Level A,C)	5.3.1.2 5.3.2.1	MIL-P-116-Visual
303. Drying inadequate or improper (Level A,C)	5.3.1.3 5.3.2.1	MIL-P-116-Visual
304. Preservative missing from unpainted surfaces, threads, or improperly applied (Level A)	5.3.1.4 5.3.1.5.2	MIL-C-11796, MIL-L-3150, MIL-P-116- Visual
305. Incorrect method of unit packaging (Level A)	5.3.1.5	MIL-P-116-Visual
306. Shield assembly not wrapped or secured as specified (level A,C)	5.3.1.5.3 5.4.1.3	Visual
307. Arm assembly not wrapped or secured as specified (Level A,C)	5.3.1.5.4 5.4.1.4	Visual
308. Drive-mount and headrest assembly not wrapped or secured as specified (Level A,C)	5.3.1.5.5 5.4.1.5	Visual
309. Water-vaporproof barrier and gaskets not applied, secured as specified, damaged or non-conforming material (Level A)	5.3.1.5.6 5.4.1.2	MIL-B-131-Visual
310. Plywood base not constructed as specified (Level A,C)	5.4.1.1 5.4.2	SME-Visual
311. Water-vaporproof bag damaged, or improperly sealed (Level A)	5.4.1.7	MIL-P-116-Visual
312. Shipping container not fabricated as specified (Level A,C)	5.4.1.12	Visual-sin
313. Strapping missing, or (Level A)	5.4.1.13	Visual

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TABLE III - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 2.5% (Cont'd)	**Req't Par.**	**Test Method**
114. Markings illegible, incorrect, missing from interior package or exterior container, non-conforming to standard(Level A,C)	5.5.1 5.5.2 5.5.3	Visual-MIL-STD-129
MINOR: AQL 4.0%		
201. Mounting hardware improperly wrapped or secured (Level A,C)	5.3.1.5.1	Visual
202. Non-conforming paper material a plied (Level A,C)	5.3.1.5.1, 5.3.1.5.2, 5.3.1.5.3, 5.3.1.5.5	MIL-B-121, MIL-P-130-Visual
203. Projections, sharp corners, edges inadequately covered or secured (Level A)	5.4.1.6	Visual
204. Link and tube assemblies improperly secured (Level A,C)	5.4.1.8 5.4.1.9	Visual
205. Protective coating on hardware, clamps, screws, missing (Level A)	5.4.1.10 5.4.1.12	Visual
206. Threads unpainted (Level A)	5.4.1.11	Visual
207. Strapping improper (Level C)	5.4.2	Visual
208. Workmanship	5.7	Visual

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TABLE IV - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 4.0%	**Req't Par.**	**Method Protection**
101. Heat seal	5.4.1.7	MIL-P-116
102. Quick leak	5.4.1.7	MIL-P-116

MINOR : NONE DEFINED

4.8 Test methods and procedures.

4.8.1 Operating temperatures.- The test shall consist of exposing the drive-mounts to the temperatures indicated in 3.5.1 by means of test chambers capable of maintaining these temperatures until the units under test are thermally stabilized. The test shall be conducted in accordance with Procedure I of Specification MIL-F-13926. While at these specified temperatures, the unit sample shall be tested for conformance to the applicable requirements of 3.5.1.

4.8.2 Vibration.- The drive-mount complete shall be vibrated as indicated in 3.5.2 utilizing a vibrating device and holding medium the accuracy of which shall conform to the requirements of 4.6.4. At the conclusion of the vibration time, the drive-mount shall be subjected to a visual and tactile examination for any evidence of damage, loose or missing parts, and for conformance to the performance requirements specified in 3.5.2.

4.8.3 Orientation.- The drive-mount shall be mounted and secured on a special testing device and oriented for inspection and testing in accordance with the requirements of 3.6.

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4.8.4 Superelevation input travel.- With the drive-mount secured and oriented as specified in 4.6.3, the input shaft shall be rotated from the clockwise stop to the counterclockwise stop, then back to the clockwise stop. During the excursion, the input shaft shall function as specified in 3.7. The minimum number of full turns of the input shaft from clockwise stop to counterclockwise stop shall be as specified in 3.7.

4.8.5 Superelevation accuracy.- This test shall be conducted optically in addition with the drive-mount positioned as in previous-test. Position gun trunnion arm at zero degrees gun elevation. With backlash excluded for counterclockwise rotation, position the input shaft at zero superelevation as indicated on the qualification plate. Affix sighting devices to the simulated gun trunnion and output coupling. Establish the line of sight through the output device parallel to the established line of sight through the gun trunnion device. The parallel lines of sight shall be directed to a target appropriate for measuring superelevation accuracy. Rotate the input shaft five complete revolutions counterclockwise from zero position and observe the rotational position of the output coupling as measured on the target. This reading shall be within the value and tolerance specified. Then continue rotation of the input shaft of ten complete revolutions in the same direction, and observe the rotational positional of the output coupling as measured on the target. This reading shall be within the value and tolerance specified in 3.8. During the performance of this test, the gun trunnion arm should be immobilized at zero degrees gun elevation.

4.8.6 Superelevation backlash.- This test will be conducted following the test in 4.85, with the designated torque applied to the output coupling. With backlash excluded for counterclockwise travel, rotate input shaft to zero superelevation and note location of the line of sight from the output coupling with reference to the target. With backlash excluded, rotate input shaft one complete turn counterclockwise, then with backlash included, rotate the input shaft one complete turn clockwise to zero and note location of the line of sight. The difference between the first and second noted reading of the line of sight is the amount of backlash in the mechanism. Repeat this test for five and ten revolutions of the input shaft from zero superelevation. Backlash resulting from rotation to any one of the three positions shall not exceed the limit specified in 3.9.1.

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4.8.7 Elevation backlash.- The designated torque shall be applied to he output coupling. The lines of sight at the gun trunnion and output coupling shall be established as in 4.8.5. Actuate gun trunnion to simulate minus one degree gun depression with an appropriate indexing device used to introduce angles into the gun trunnion. With backlash excluded, elevate gun trunnion to zero degree gun elevation and record location of the line of sight. Actuate gun trunnion to simulate plus one degree gun elevation. With backlash excluded, depress gun trunnion to zero degree gun depression and record location of the line of sight. The line of sight deviation at the output coupling shall be measured as the difference between the two recorded readings, and this reading shall not deviate in excess of the limit specified. After recording the line of sight readings at the eight degree gun elevation position, elevate the gun trunnion to exclude backlash and return to eight degree gun elevation position and record line of sight readings after recording line of sight readings at fifteen degrees gun elevation position, elevate the gun trunnion to exclude backlash and return to fifteen degrees gun elevation position and record line of sight readings. The difference between the first and second readings at 8 and 15 degrees simulated gun positions is the elevation backlash. This backlash shall not exceed the limit specified in 3.9.2.

4.8.8 Shield assembly.- When the plunger in the handle assembly is depressed, the handle is elevated to the uppermost position and the plunger released to engage in its detent, the shield shall be secured in the open position. When the plunger is depressed, the handle moved to the lowermost position, and the plunger released to engage in its detent, the shield shall be secured in the closed position.

4.8.9 Release grip.- The function of the lever assembly shall be tested by inserting an acceptable Periscope M29, or the equipvalent thereof, into the drive-mount. The lever assembly shall automatically engage and retain the periscope as specified in 3.11.1.

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4.8.9.1 The force required to disengage the lever to free the periscope shall not exceed the value specified. The force shall be applied, using standard measuring equipment, to the lever and shall be in a plane parallel to Shaft 8298190. When the lever is pulled to the rearmost position, the periscope shall be withdrawn from the drive-mount without restriction.

4.8.9.2 Lever.- The test shall be conducted by use of the testing device capable of measuring the required pressure. The Periscope M29, or its equivalent thereof, shall be inserted in the drive-mount and the lever engaged in its forward position. When the required pressure is applied, front and rear in turn, to the lowest point of the body assembly of the testing periscope in the drive-mount. The line of sight emerging from the opposite end of the periscope shall not deviate in any plane more than the limit specified in 3.11.2. This movement of the periscope line of sight shall be observed and measured optically with reference to an appropriate target.

4.8.10 Headrest.- The headrest shall be manually operated left and right on its frame and shall function as specified in 3.12.

4.8.11 Spindle arm.- The test shall be conducted using a commercial type torque device with an adapter to accept the spindle of the arm assembly. The device shall be affixed to the spindle. The torque required to rotate the spindle through the range of objective travel at standard ambient temperature shall be measured and recorded. This reading shall not exceed the value specified in 3.13.1.

4.8.12 Eccentric.- The test shall be conducted using a commercial type torque device with an adapter to accept the spindle eccentric. The device shall be affixed to the eccentric. The torque required to rotate the eccentric through the range of objective travel at standard ambient temperature shall be measured. This reading shall not exceed the value specified in 3.13.2.

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4.8.13 Identification marking.- The drive-mount shall be visually checked for proper identification. Marking shall be visually checked for conformance to Standard MIL-STD-130.

4.8.14 Preproduction pack.- The preproduction pack shall be inspected by the contractor and witnessed by the Government representative to determine conformance with the requirements of 5.6.

4.8.15 Rough handling test.- The preproduction pack only shall be subjected by the rough handling test in accordance with the procedures outlined in Specification MIL-P-116.

4.8.16 Preproduction pack failure.- Should the preproduction pack fail to pass he acceptance tests, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be selected and completely inspected by the contractor and witnessed by the Government representative.

4.8.17 Packaging tests.- Each package in the applicable sample shall be subjected to the pertinent packaging test of Specification MIL-P-116 as determined by the method of protection shown in the Table IV.

4.8.18 Determination of cleanliness test. Periodically, the applicable surfaces of one or more units shall be subjected to the "Determination of Cleanliness" test in accordance with Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 Materials and methods.- The materials, methods and procedures used in he packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

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5.1.1 Materials.- The following shall apply wherever packaging materials are referred to in its specification by name only:

REFERENCE	FULL IDENTIFICATION
Filament tape	Type III, at least 3/4 inch wide, of Specification PPP-T-97
Greaseproofed barrier material	Type I or II, grade A, class 2 of Specification MIL-B-121
Paperboard cushioning material	Type I, class 1 of Specification PPP-P-291
Pressure sensitive tape	Type I of Specification UU-T-106
Paper material	Type I of Specification MIL-P-130
Waterproof pressure sensitive tape	Type II, class 1 of Specification PPP-T-60

5.2 Size of containers.- The floating bag shall be of a minimum size that will adequately contain the items.

5.3 Preservation and packaging.

5.3.1 Level A.

5.3.1.1 Disassembling

5.3.1.1.1 Link assembly.- The arm assembly shall be disassembled from the link assembly. The stud shall not be removed from the link assembly. The link assembly shall then be disassembled from the connector assembly.

5.3.1.1.2 Temperature compensating tube assembly.- The stud shall be removed from the sleeve and the tube assembly. The sleeve shall then be removed from the housing assembly.

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5.3.1.1.3 Headrest.- The headrest assembly shall be removed from the drive-mount. The screws and washers, which secure the headrest assembly to the drive-mount, shall be replaced in position in the drive-mount.

5.3.1.2 Cleaning. - Cleaning Process C-1 of Specification MIL-P-116 shall followed to clean the surfaces of the drive-mount and the mounting hardware.

5.3.1.3 Drying.- Immediately after cleaning, the surfaces shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the item.

5.3.1.4 Preservation application.- The unpainted surfaces of the link assembly and tube assembly shall be coated with preservative conforming to class 2 of Specification MIL-C-11796 (P-6 of Specification MIL-P-116). The unpainted metal surfaces of the shield assembly, connector assembly, arm assembly, sleeve and stud shall be coated with a thin film of preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116). Excess preservative shall be removed.

5.3.1.5 Unit packaging. - Packaging the drive-mount shall be as specified herein is in conformance with Method IA-16 of Specification MIL-P-116. Packaging the link assembly and tube assembly shall be as specified herein which is in conformance with Method I of Specification MIL-P-116.

5.3.1.5.1 Mounting hardware.- The mounting hardware for the mount shall be placed together and wrapped with paper material. The wrap shall be secured in place with pressure sensitive tape. The wrapped hardware for the mount and bracket shall then be secured to the drive-mount with filament tape. The wrapped hardware for the mount and the bracket shall then be secured to the drive-mount with filament tape.

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5.3.1.5.2 Covering preservative coated surfaces and mounting surfaces.- The preservative coated surfaces of the link assembly, (threaded end and edn with the stud) tube assembly, (open-ends) sleeve and stud shall be wrapped with greaseproofed barrier matetial. The wrapped sleeve and stud shall be placed together and wrapped with paper material. The wrap shall be secured in place with pressure sensitive tape. The wrapped sleeve and stud shall be secured to the drive-mount with filament tape. The wrapped ends of the link assembly and the tube assembly shall then be over-wrapped with waterproof pressure sensitive tape in such a manner that each end of the link assembly and the tube assembly and approximately 1 inch of painted surfaces beyond greaseproofed barrier material are covered. The mounting surfaces of the shield assembly shall be covered with greaseproofed barrier material. Bolt holes shall be provided in the greaseproofed barrier material for bolts to pass through for securing the shield assembly to the plywood base shown in Figure 1 and 3.

5.3.1.5.3 Shield assembly.- The shield and the fork assembly shall be secured to the cover with filament tape (shown in Figures 3 and 5) in such a manner that the fork does not contact the water-vaporproof barrier material that covers the plywood base when the shield assembly is bolted to the base, specified in 5.4.1.3.

5.3.1.5.4 Arm assembly.- The portion of the arm assembly over which the metal clamp (shown in Figures 4 and 5) is to be placed, specified in 5.4.1.4, shall be wrapped with paperboard cushioning material. The wrap shall be secured in place with pressure sensitive tape.

5.3.1.5.5 Drive-mount and headrest assembly.- The metal cover on the level vial assembly shall be positioned in such a manner that the vial is covered. pressure sensitive tape shall be wrapped around the vial. The surface of the connector assembly over which the metal clamp (shown in Figures 4 and 5) is to be placed, specified in 5.4.1.5, shall be wrapped with paperboard cushioning. The wrap shall be secured in place with pressure sensitive tape. The two handles shall be wrapped with paperboard cushioning material as shown in Figures 3 and 5. The wraps shall be secured in place with pressure sensitive tape. The headrest assembly shall be wrapped with paper barrier material. The wrap shall be secured in place with pressure sensitive tape. The wrapped headrest assembly shall then be placed inside the mount as shown Figure 5, and secured in place with filament tape.

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5.3.1.5.6 Water-vaporproof barrier and gaskets.- Water-vaporproof barrier material, conforming to class Specification MIL-B-131, shall be used in fabricating the floating bag specified in 5.4.1.7. Appropriate size bolt holes shall be made in the barrier material to permit the bolts to pass through for securing the items to the plywood base, specified in 5.4.1.1. Gaskets with appropriate size bolt holes for the bolts to pass through shall be cemented to each side of the water-vaporproof barrier in such a manner that the holes in the gaskets and the holes in the water-vaporproof barrier coincide. The gaskets shall be fabricated from material not less than 3/32 of an inch thick and conforming to class II, grade 60 of Specification MIL-R-6855. The adhesive used to cement the gaskets in place shall conform to type III of Specification MIL-A-5092. The length and width of the gaskets shall be such that they will cover the entire mounting surfaces of the shield assembly and drive-mount, shown in Figure 4, also, the plywood pieces, shown in Figures 1 and 4, on which the arm assembly and connector assembly rest, specified in 5.4.1.4 and 5.4.1.5.

5.3.2 Level C.

5.3.2.1 Disassembling, cleaning and drying.- Disassembling, cleaning and drying shall be as specified in 5.3.1.1, 5.3.1.2 and 5.3.1.3.

5.3.2.2 Preservative application.- Preservative application shall be as specified in 5.3.1.4, except that the unpainted surfaces of the link assembly and tube-assembly shall be coated with preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116) in lieu of the preservative specified in 5.3.1.4.

5.3.2.3 Unit packages.- The drive-mount shall be packaged as specified in5.3.1.5.1 to 5.3.1.5.5 inclusive, which is in conformance with Method I of Specification MIL-P-116.

5.4 Packing.

5.4.1 Level A.

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5.4.1.1 Plywood base.- The plywood base to which the items are bolted and secured shall be constructed as shown in Figure 1.

5.4.1.2 Securing water-vaporproof barrier and gaskets to plywood base.- The adhesive, specified in 5.3.l.5.6 shall be applied on the surfaces of the gaskets which will contact the plywood base prior to placing the water-vaporproof barrier on the base. The water-vaporproof barrier shall then be positioned on the base in such a manner that the bolt holes in the base, the barrier and gaskets coincide.

5.4.1.3 Securing shield assembly to plywood base.- The shield assembly shall be positioned on the gaskets and secured to the base with two 1/2 inch step bolts, flat washers, lock washers and nuts.

5.4.1.4 Securing arm assembly to plywood base.- The plywood piece, shown in Figure 1, shall be positioned on the gasket, and the arm assembly shall be positioned on the plywood piece shown in Figure 5, the metal clamp, shown in Figure 1, shall then be placed over the wrapped surface of the arm assembly. Two 1/4 inch step bolts, lock washers and nuts shall be utilized to secure the metal strap and arm assembly to the base as shown in Figure 5.

5.4.1.5 Securing drive-mount to plywood base.- The drive-mount shall be positioned on the gasket and secured to the base with four 1/2 inch step bolts, flat washers, lock washers and nuts. The plywood piece, shown in Figure 1, shall be positioned on the gasket beneath the connector assembly. The metal clamp, shown in Figure 1, shall be placed over the wrapped surface of the connector assembly as shown in Figure 5. Two 3/8 inch step bolts, lock washers and nuts shall be utilized to secure the metal strap and assemblies-to the plywood base.

5.4.1.6 Covering prejections, sharp corners and edges.- Paperboard cushioning material shall be used to cover projections, sharp corners and edges that would puncture the bag specified in 5.4.1.7. The paperboard cushioning material shall be scoured in place with pressure sensitive tape as shown in Figure 6.

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5.4.1.7 Water-vaporproof bag.- The water-vaporproof barrier material shall be heat sealed in such a manner as to form a bag that will pass the heat seal and quick leak tests of Specification MIL-P-116. Entrapped air shall be exhausted prior to final closure.

5.4.1.8 Securing link assembly.- A 3/8 by 16 inch long stud shall be secured to each header on the plywood base with flat washers, lock washers and nuts as shown in Figures 4 and 5 A nut shall be placed on the free end of each stud-and screwed down to the end of the threads. Metal clamps (shown in Figures 1 and 6) for the large and small diameters of the link assembly shall be positioned on the assembly. The clamps, with the tube assembly, shall then be placed on top of the nuts. A lock washer and a nut shall be added on each stud above the clamps. The nuts shall then be tightened until the clamps secure the link assembly (see Figure 7).

5.4.1.9 Securing tube assembly.- The tube assembly shall be secured to the plywood base with two 1 inch pipe straps, bolts, Lock washers and nuts. The tube assembly shall be located on the base in such a manner that the ends of the tube are at least 1 inch from the sides or ends of the container.

5.4.1.10 Protective coating on hardware and clamps.- All clamps, bolts, studs, flat washers, lock washers and nuts, used in packing the items, shall be either zinc or cadmium coated.

5.4.1.11 Applying Unthinned paint.- Unthinned paint shall be applied on all exposed threads above nuts to prevent the nuts from backing off.

5.4.1.12 Sides, ends and top panels of shipping container.- The sides, ends and top panels of the shipping container shall be constructed as shown in Figure 2. The top panel shall be secured in place with zinc coated steel flat head wood screws (no. 8 by 1-1/4 inches long). The screws shall be spaced approximately four inches apart.

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5.4.1.13 Strapping.- The shipping container shall be strapped lengtwise with two metal straps in such a reamer that each strap is adjacent to each skid and approximately parallel with the skid. The straps, staples and stapling method shall be in accordance with Specification PPP-B-621.

5.4.2 Level C.- Packing shall be as specified in 5.4.1.1 to 5.4.1.13 inclusive except as follows:

(a) Adhesive, gaskets, water-vaporproof barrier material and water-vaporproof bag, specified in 5.4.1.2 to 5.4.1.7 inclusive, shall be omitted.

(b) The plywood pieces specified in 5.4.1.4, 5.4.1.5 and Figure 1, shall be omitted.

(c) Covering projections, sharp cormers and edges specified in 5.4.1.6, shall be omitted.

(d) The protective coating on hardware and screws, specified in 5.4.1.10 and 5.4.1.12, shall be omitted.

(e) The metal strapping on the shipping container shall not be stapled.

5.5 Marking.

5.5.1 Interior packages.- Marking of interior packages shall be in accordance with Standard MIL-STD-129. The serial number shall be added in the identification data on each level A unit package and omitted on the Level C unit package.

5.5.2 Shipping container. - Each shipping container shall be marked in accordance with standard MIL-STD-129. The serial number of the drive-mount in a Level A shipping container shall be added in the identification data and omitted on the Level C shipping container.

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5.5.3 Stock number.- The stock number to be added in the identification data shall be obtained from the contracting officer.

5.6 Preproduction pack.- Prior to quantity packaging production, a preproduction pack on pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification A rough handling test is required.

5.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

6. **NOTES**

6.1. Intended use.- The drive-mount, primarily intended for use with the 120MM Gun Tank M102A1, is a combination ballistics drive and periscope mount. This drive-mount maintains elevation synchronization between the lines of sight of the gun and the gunner's periscope and introduces superelevation into the periscope as received from a ballistics computer.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Size and weight of packed shipping container. - The following outside dimenslons, overall cubage and grass weight are approximate for one periscope drive-mount packaged and packed for Level A:

Outside dimensions - 38-1/2 inches long by 28. inches wide by 17-1/2 inches deep
Cubage - 11. cubic feet
Gross weight - 275. pounds

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6.4 Definitions.

6.4.1 Tactile.- Skillful manipulation or sensing.

6.4.2 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments and modified-standard measuring equipment.

6.4.3 Standard measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial-supply houses for ready supply (shelf items), and which are normally used by an inspector to perfom dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.5 Design infomation.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.6 Drawings.- All drawings referenced throughout Section 3 are components of the drawing set F8606140 Drive-mount, Periscope: M9 and drawings pertaining thereto.

6.7 Performance testing.- While the drive-mount is undergoing performance testing Screw MS 35234-45, Screw MS 35234-48 (with washers), Pin 583141 and Pin 544151 shall be removed. After final acceptance, the pins and screws shall be inserted prior to shipment and shall only be removed in accordance with the note specified on Drawing F8606140.

NOTICE : When Government drawings, specifications, or other data are used for any prupose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

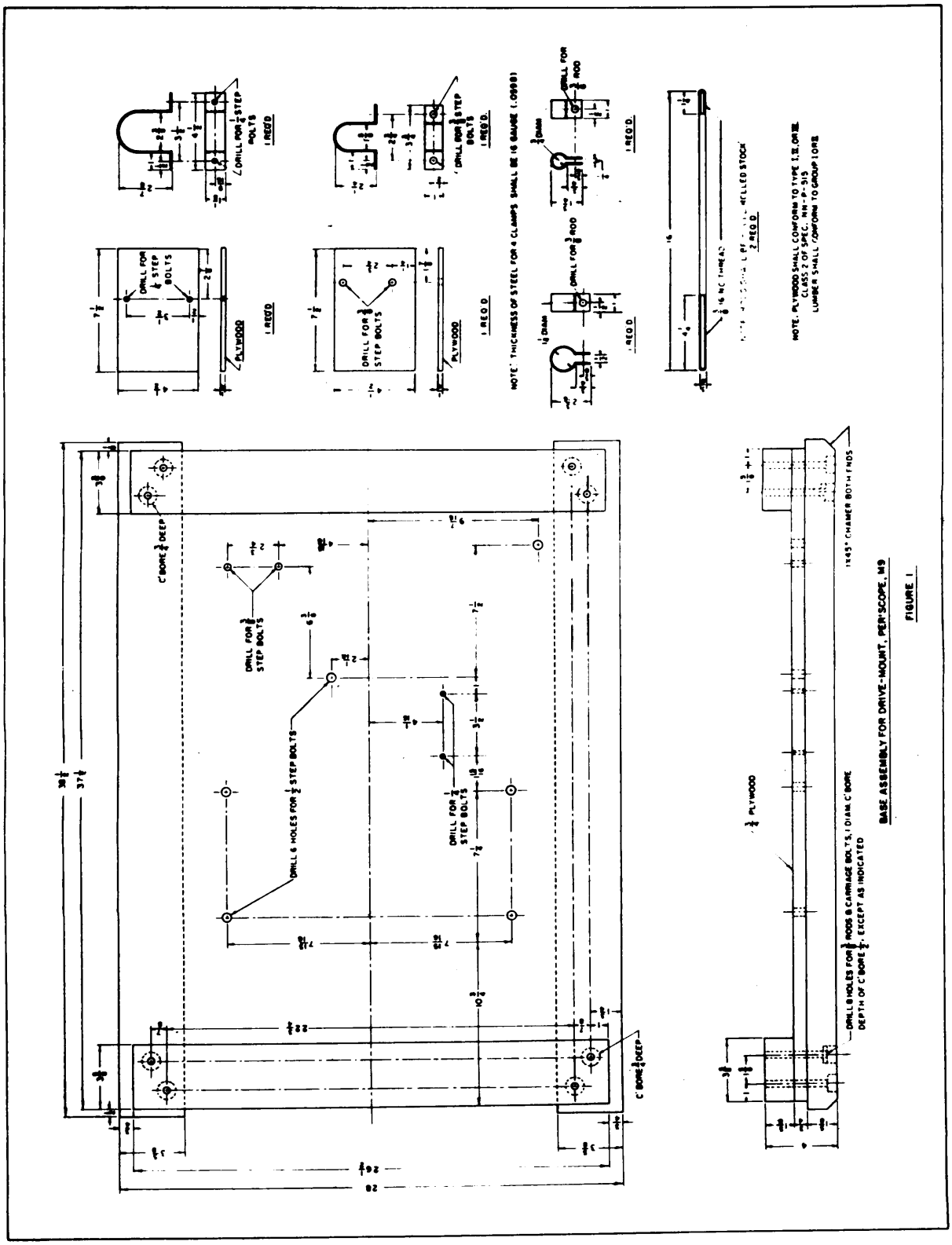
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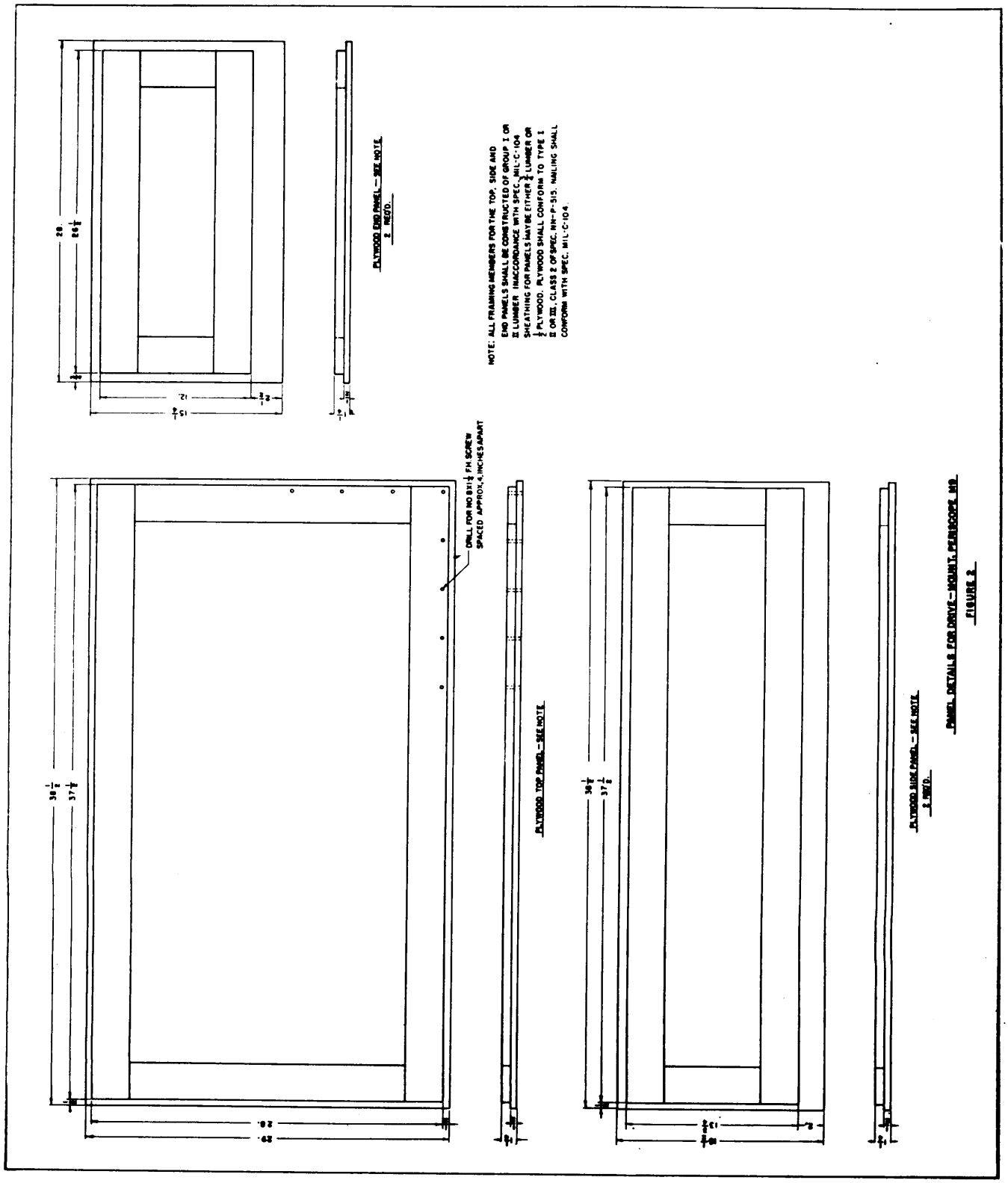
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BASE ASSEMBLY FOR DRIVE-MOUNT, PERISCOPE, M9

FIGURE 1

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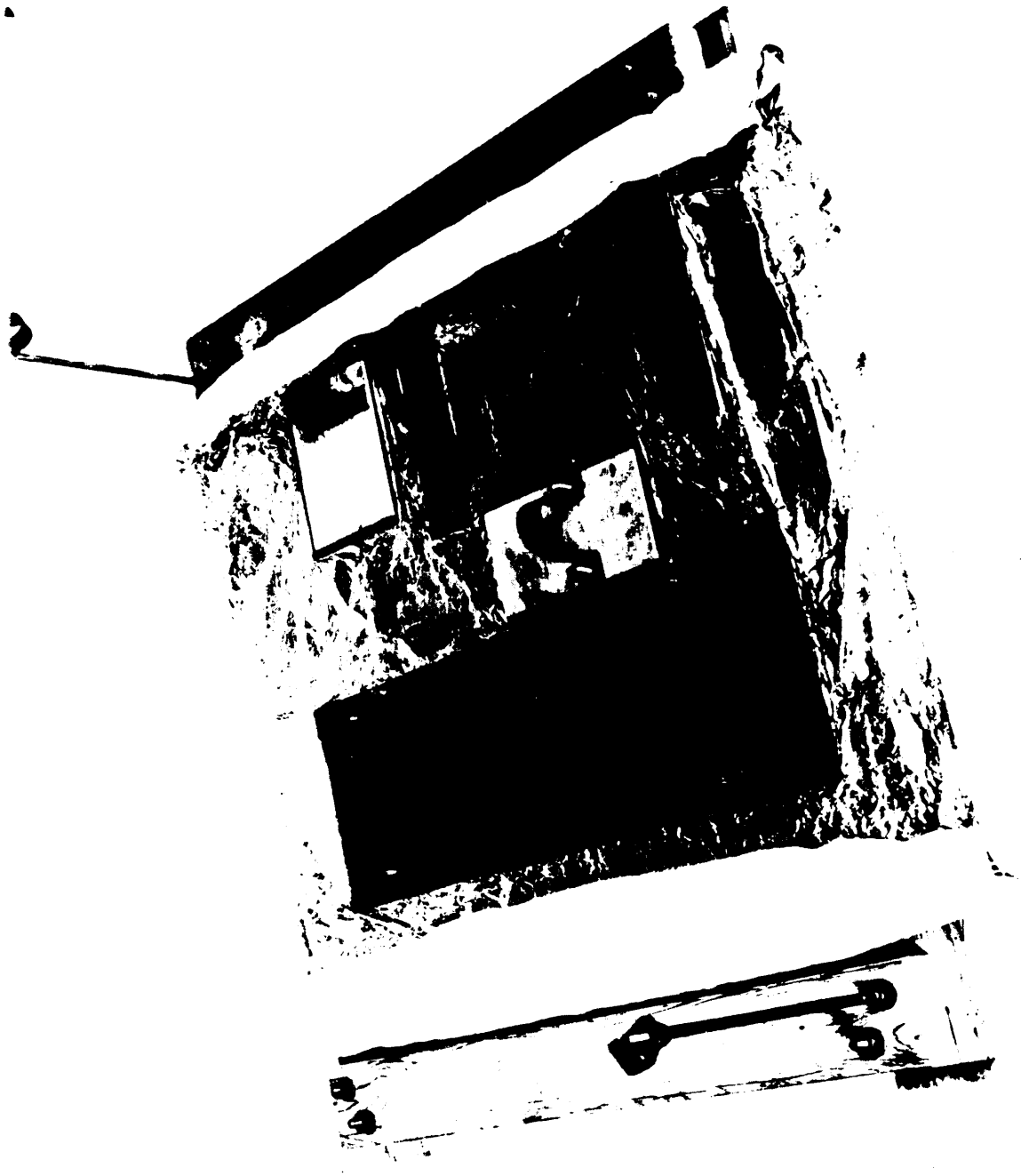
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DRIVE-MOUNT, PERISCOPE, M9 - Showing wrapped hardware, headrest and level vial; immobilized fork assembly; greaseproofed barrier material on mounting surfaces of shield assembly; wrapped surfaces on arm and connector assemblies; temperature compensating tube assembly not shown.

FIGURE 3

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DRIVE-MOUNT, PERISCOPE, M9 - Showing base, waterproof barrier, gaskets, plywood pieces, metal clamps, bolts and studs.

FIGURE 4

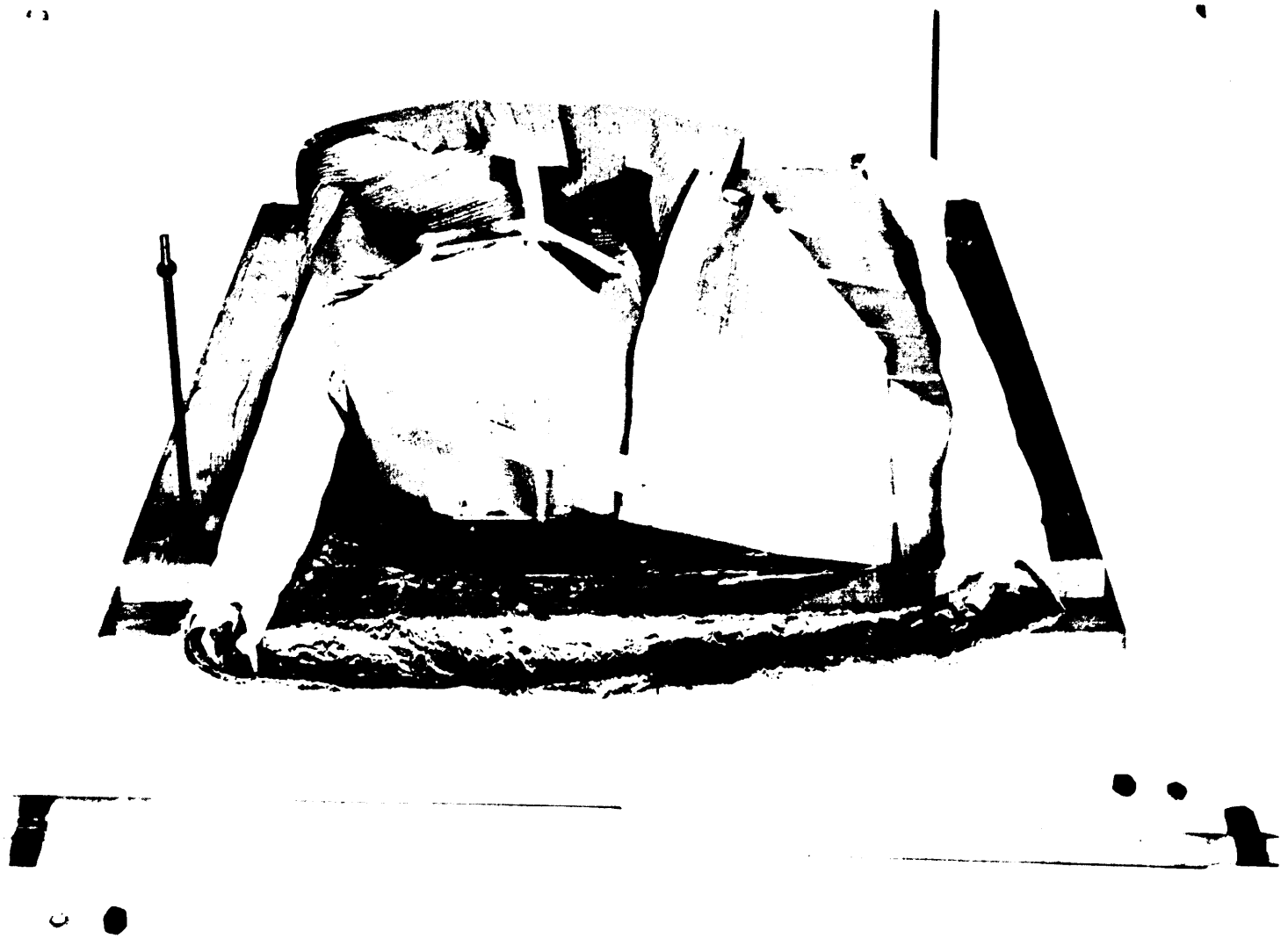
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DRIVE-MOUNT, PERISCOPE, M9 - Showing wrapped hardware and headrest assembly secured to drive-mount, wrapped handles, and drive-mount with loose items secured to water-vaporproof barrier covered plywood base.

FIGURE 5

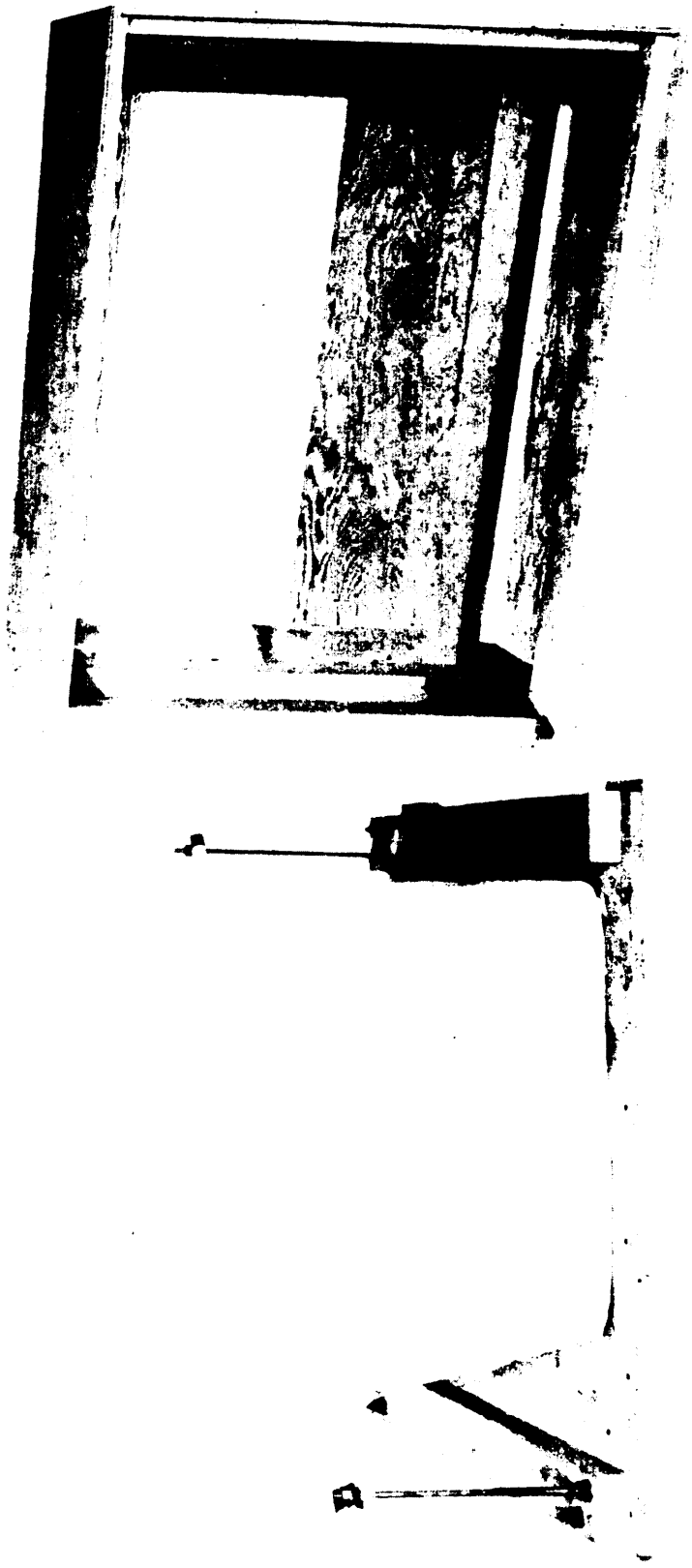
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DRIVE-MOUNT, PERISCOPE, M9 - Showing projections, sharp edges and corners covered prior to heat sealing the water-vaporproof barrier; also, link assembly prior to positioning and securing on studs; temperature compensating tube assembly not shown

FIGURE 6

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DRIVE-MOUNT, PERISCOPE, M9 - Drive-mount packaged and packed prior to securing sides, ends and top panels in place.

FIGURE 7

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